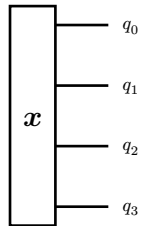
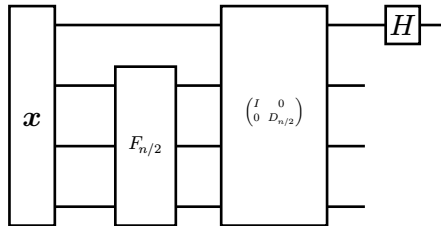


(a) Re-index

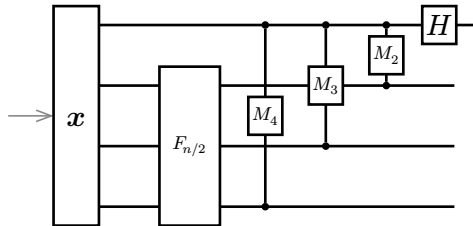
 x as rank- k tensor

(b) Recurse

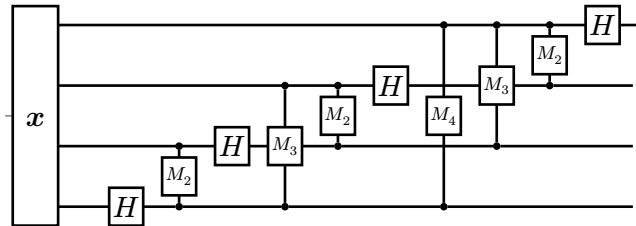


one butterfly layer

(c) Decompose

twiddle $\rightarrow M_j$ chain

(d) Relax

 $H \in U(2), \quad M_j \in U(1)^4$